

ANALYSIS AND DESIGN OF ALGORITHMS

Semester	IV				
Course code					
Category	Professional Core Courses				
Course title	Analysis and design of Algorithms				
Scheme and Credits	L	T	P	Credits	
	3	0	0	3	
Classwork	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03 Hours				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVE:

1. The students should be able to analyze various algorithms mainly for time and space complexity.
2. They should be able to develop algorithm for solving various computational problems by applying various algorithm design strategies.
3. They should be able to understand the effect of choice of data structures on the complexity of algorithm

UNIT I

BASIC CONCEPTS OF ALGORITHMS: Basic Concepts of Algorithms: Notion of Algorithm, Fundamentals of Algorithmic Solving, Important problem types, Fundamentals of the Analysis Framework, Asymptotic Notations and Basic Efficiency Classes, Mathematical analysis of non-recursive algorithms. Mathematical analysis of recursive algorithm: recurrence relations, solution of recurrence relations using substitution method

UNIT II

SORTING ALGORITHMS: Brute Force, Divide and Conquer Strategy: Selection sort, Bubble sort, Sequential searching (Linear Search), Brute force string matching, General method, Merge sort, Quick Sort, Binary Search, Strassen's matrix multiplication

UNIT III

GREEDY AND DYNAMIC PROGRAMMING: Greedy Approach and Dynamic Programming: Fractional Knapsack problem, Minimum cost spanning tree: Prim's and Kruskal's algorithm, Single source shortest path problem, Principle of optimality, Multi-stage graph problem, all pair shortest path problem, 0/1 Knapsack problem, Traveling salesperson problem

UNIT IV

BACKTRACKING & BRANCH AND BOUND: Backtracking and Branch and Bound: General

method backtracking, N-Queen problem, 0/1 Knapsack problem, General method of branch & bound, 0/1 Knapsack problem, Traveling sales person problem
Lower Bound Theory and Complexity Classes, Lower bounds, Decision trees, P, NP and NP Complete problems

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

- CO1: Apply the best data structure for designing an algorithm to solve a given problem.
- CO2: Evaluate different algorithms with respect to time and space complexity.
- CO3: Create algorithms to solve various computational problems.
- CO4: Understand different complexity classes.
- CO5: Understand the sorting techniques
- CO6: Explain the hardness of real world problems with respect to algorithmic efficiency and learning to cope with it.

TEXT AND REFERENCE BOOKS:

1. Algorithm Design, Jon Kellinberg and Eva Tardos, 1st Edition, Pearson Education 2014.
2. Design & Analysis of Algorithms, Gajendra Sharma, Khanna Book Publishing 2018.
3. Fundamentals of algorithms, Horowitz E, Sahini S, Rajasekaran S., University Press 2008.
4. Introduction to algorithms, Cormen, Leiserson, Rivest, Stein, 3rd Edition, PHI. 2012
5. An introduction to analysis of algorithms, R. Sedgewick, 1st edition, Pearson Education 1996.
6. Data Structures and Program Design in C By Robert L. Kruse, C.L. Tondo, Bruce Leung, Pearson Education. 2007.